Tarkan Zarrouk

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Education

Carleton University — Ottawa, Ontario
Honours Bachelor of Mathematics — *Sep. 2025 – Present*

- Relevant coursework: Data Structures and Algorithms, Discrete Mathematics, Calculus, and Computer Systems.
- Recipient of the Presidential Entrance Scholarship valued at $6,000.

High School — Ontario, Canada
Ontario Secondary School Diploma — *Sep. 2021 – Jun. 2025*

- Graduated with an 88.75% cumulative average.
- Achieved 98% in Grade 12 Computer Science and 96% in Computer Engineering Technology.
- Recognized for the highest overall average in Computer Engineering Technology.

Experience

Frontend Web Developer — Feb. 2025 – Apr. 2025
Kaie — *Mississauga, Ontario*

- Developed and improved user-facing web application features, including user profile management and an Explore page, to improve personalization and content discoverability.
- Implemented frontend components and integrated application functionality with backend services and data structures.
- Contributed to the migration of 2,000+ user accounts to a new database while maintaining data integrity and minimizing service disruption.

Code Sensei — Jul. 2024 – Aug. 2024
Code Ninjas — *Ontario, Canada*

- Taught programming concepts to children aged 6–13 through hands-on coding activities and classroom instruction
- Guided students through object-oriented programming, algorithms, debugging, and problem-solving.
- Collaborated with colleagues to deliver age-appropriate lessons and maintain an engaging learning environment.

Projects

Social Media Application | *React, TypeScript, Firebase, JavaScript* — Feb. 2025 – Apr. 2025

- Built a database-driven social media application to develop practical experience with CRUD operations, application architecture, and backend integration.
- Integrated Firebase services to manage application data and support user-facing functionality.
- Implemented data handling and validation to improve reliability when creating, retrieving, updating, and deleting application records.

Computer Vision Detection System | *Python, OpenCV, YOLOv5*

- Developed computer vision prototypes for face and QR code detection using YOLOv5 and OpenCV.
- Worked with trained model weights and image processing pipelines to detect and classify visual features.
- Explored object detection and computer vision techniques for real-world recognition tasks.

Real-Time Tracking System | *Python, MediaPipe, Arduino*

- Experimented with MediaPipe-based tracking and integrated visual input with Arduino-controlled hardware.
- Connected software processing with physical actuator control to explore interactive tracking systems.

Technical Skills

Languages: Python, R, MATLAB, SmallTalk, Java, C, C++, JavaScript, TypeScript, HTML, CSS, SQL
Frameworks and Technologies: Jupyter Notebook, React, Node.js, Firebase, OpenCV, YOLOv5, MediaPipe, Arduino
Developer Tools: Git, GitHub, VS Code, IntelliJ IDEA, Linux

Awards and Certifications

Presidential Entrance Scholarship — 2025
Carleton University